PYTHON PROGRAMMING



21 Days Challenge

Day-07



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PYTHON DICTIONARY ESSENTIALS: CREATION, ACCESS, MODIFICATION & NESTING

Instructor



Abdullah Al Zubaer

Instructor – Artificial Intelligence & Data Science

B.Sc. in Internet of Things and Robotics Engineering

Gazipur Digital University, Gazipur, Bangladesh

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Dictionary

A Python dictionary is a built-in data structure that stores data in key-value pairs.

```
STUDENT = {
    "NAME": "ALICE",
    "AGE": 21,
    "GRADES": [85, 90, 92],
    ("YEAR", 2025): "GRADUATION YEAR"
}
```

Properties of Python Dictionary

- Keys must be immutable (strings, numbers, or tuples) and unique.
- Values can be of any data type and can be duplicated.
- Dictionaries are ordered mutable, and indexed by keys.

```
STUDENT = {
    "NAME": "ALICE",
    "AGE": 21,
    "GRADES": [85, 90, 92],
    ("YEAR", 2025): "GRADUATION YEAR",
    #[89, 95, 97]:"MARKS",# ERROR
    "AGE": 29
}
```

Output

```
{'name': 'Alice', 'age': 29,
'grades': [85, 90, 92],
('year', 2025): 'Graduation Year'}
```

Creating Dictionary

```
sports_player =
{
"Name": "Sachin Tendulkar",
"Age": 48,
"Sport": "Cricket"
}

print (sports_player)
```

Output

```
{'Name': 'Sachin Tendulkar', 'Age': 48, 'Sport': 'Cricket'}
```

Creating Dictionary

```
# Creating a dictionary using the dict() function

student_info = dict(name="Alice", age=21, major="Computer Science")

print(student_info)
```

Output

```
{'Name': 'Sachin Tendulkar', 'Age': 48, 'Sport': 'Cricket'}
```

Accessing Dictionary

```
person = {"name": "Alice", "age": 25}
print(person["name"])
print(person.get("name"))
```

Output

```
Alice
Alice
```

Modifying Dictionary

```
student_info = {
"name": "Alice",
"age": 21,
"major": "Computer Science"
}
student_info["age"] = 22 # modifying
student_info["graduation_year"] = 2023 # new key-value adding
print(student_info)
```

Output

```
{'name': 'Alice', 'age': 22, 'major': 'Computer Science',
'graduation_year': 2023}
```

Iterating Through a Dictionary

```
student_info = {
"name": "Alice",
"age": 21,
"major": "Computer Science"
}
for key, value in student_info.items():
    print(key,":", value)
```

Output

```
name : Alice
age : 21
major : Computer Science
```

Nested Dictionaries

```
nested_dict = {
"box1": {"item1": "apple", "item2": "banana"},
"box2": {"item1": "carrot", "item2": "dates"}
}
print(nested_dict)
```

Output

```
{
'box1': {'item1': 'apple', 'item2': 'banana'},
'box2': {'item1': 'carrot', 'item2': 'dates'}
}
```

Nested Dictionaries

```
nested_dict = {}
outer_keys = ["box1", "box2"]


for key in outer_keys:
nested_dict[key] = {"item1": "apple", "item2": "banana"}
print(nested_dict)
```

Output

```
{
'box1': {'item1': 'apple', 'item2': 'banana'},
'box2': {'item1': 'carrot', 'item2': 'dates'}
}
```

Adding in Nested Dictionaries

```
boxes = {
    "box1": {"item1": "apple", "item2": "banana"},
    "box2": {"item1": "carrot", "item2": "dates"}
}
boxes["box1"]["item3"] = "egg" # adding
print(boxes)
del boxes["box1"]["item2"]  # deleting
print(boxes)
```

Output

```
{'box1': {'item1': 'apple', 'item2': 'banana',
'item3': 'egg'}}
{'box1': {'item1': 'apple', 'item3': 'egg'}}
```

Iterating Nested Dictionaries

```
students = {
    "student1": {"name": "Alice", "age": 21, "major":
"Computer Science"},
}

for student_key, details in students.items():
    print(student_key)
    for key, value in details.items():
        print(f" {key}: {value}")
```

Output

```
student1
 name: Alice
 age: 21
 major: Computer Science
```

QUESTION SESSION

THANK YOU

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